CSE 6550 Software Engineering

Local Civic Issue Tracker

Software Requirements Specification v1.0

Table 1. Document control

Document control	Entry
Project title	Local Civic Issue Tracker
Document identifier	LCIT-SRS
Version and status	1.0 | Draft for team review
Date	October 30, 2026
Requirements owner	Kazim Yigit Ablay and Michael Avila
Team members	Kazim Yigit Ablay and Michael Avila
Primary audience	Project team, instructor and course reviewers
Basis	CSE 6550 SRS template, course requirements and agreed project scope

This specification records the agreed project scope and requirements for design, implementation and verification. Conditional features and unresolved items are identified explicitly, and planned verification is distinguished from completed testing.

Revision History

Table 2. Revision history

Version	Date	Author or owner	Summary
0.1	Oct 5, 2026	Kazim Yigit Ablay	Partial draft for joint completion.
0.2	Oct 5, 2026	Michael Avila	Working draft pending joint review.
1.0	October 30, 2026	Kazim Yigit Ablay	Expanded and refined the v0.2 requirements to reflect agreed scope, processing, data, quality and security decisions. Updated constraints, verification, traceability and supporting records.

1 Introduction

1.1 Purpose

This SRS specifies the current requirements for Local Civic Issue Tracker. It guides design, implementation, verification and controlled change. Version 1.0 expands the submitted v0.2 baseline while keeping conditional features and unresolved acceptance details explicit.

1.2 Product Scope

Following a local issue can require reading documents from several meetings, even when those documents describe the same project or concern. Local Civic Issue Tracker will bring related public records together so users can follow an issue and consult the original information. The initial scope is San Bernardino City Council records from 2023 onward, with a smaller set used for the project demonstration. The product includes issue timelines, meeting overviews, and summaries supported by official records.

The product reconstructs the currently supported history of an issue using selected official records now available. Later records may establish earlier developments. An issue timeline follows the evolution of a specific civic matter, from its first supported mention in the imported collection to its latest supported state.

1.3 System Boundary and Non Goals

The system imports selected official records, including existing transcripts and cancellation notices; searches issues, records and meetings; identifies supported connections; and presents issue timelines, meeting overviews, summaries and original-source links. Links to official meeting videos are included when available. Broad issue categorization is conditional on a simple implementation within existing analysis work.

Other cities, additional boards and commissions, pre-2023 archives, and 3-1-1 integration are outside the initial scope. Generating transcripts and analyzing video content are also outside this scope. Pictures, maps, and further reading links remain stretch goals. Political advice and automated civic decision-making are outside the project scope.

Non-English entry processing is not mandatory. The baseline does not provide historical as-known-then replay. Extracting text from scanned pages or source graphics does not bring map or picture presentation into the required feature scope.

1.4 Intended Audience and Use

Kazim Yigit Ablay and Michael Avila will use this document to develop and review the project. The instructor and course reviewers will use it to assess the requirements and the resulting work.

1.5 References

Table 3. References

ID	Reference	Use
REF-01	CSE 6550 Project SRS v0.1 Requirements Baseline	Assignment requirements
REF-02	CSE 6550 SRS v0.1 Requirements Baseline Template	Document structure
REF-03	ISO/IEC/IEEE 29148:2018, Systems and software engineering - Life cycle processes - Requirements engineering	SRS and requirement-quality guidance
REF-04	Kazim Yigit Ablay, Local Civic Issue Tracker project pitch and speaker notes	Original project purpose and proposed direction
REF-05	Trailhead SRS ISO 29148 Aligned worked example	Structure and level of detail
REF-06	City of San Bernardino, Agendas & Minutes https://www.sanbernardino.gov/720/Agendas-Minutes	Official record-access page
REF-07	Google Java Style Guide https://google.github.io/styleguide/javaguide.html	Selected Java coding-style basis

ID	Reference	Use
REF-08	Alistair Mavin, EARS official guide https://alistairmavin.com/ears/	Requirement sentence patterns
REF-09	PR-01 team review handoff and retained experiment records, October 11, 2026, Kazim Yigit Ablay	Selected-record acquisition/extraction feasibility and its limits
REF-10	CSE 6550 Software Requirements Specification update assignment	EARS, measurable quality, security, revision and traceability expectations

1.6 Terms and Abbreviations

Table 4. Terms and abbreviations

Term	Definition
SRS	Software Requirements Specification
MVC	Model View Controller
AI	Artificial intelligence
Record	An official municipal document or agenda item used by the application.
Meeting overview	A summary of a meeting's agenda topics and recorded outcomes.
Verification	Checking whether the system satisfies its specified requirements.
Validation	Checking whether the requirements and system address the intended needs.
Issue	A specific ongoing civic matter, such as a project, proposal or policy change, whose developments can be followed across meetings. A broad topic alone does not establish issue identity.
Development	A source-supported proposal, action, occurrence or change, distinct from its evidence and ongoing issue.
Source evidence	An identifiable passage in an official source that supports a retained claim or relationship.
Source-stated / inferred connection	A relationship explicitly stated by an official source / an association established by the system using supporting context.
Expected input	A portal-listed record within the agreed import scope for a selected meeting.
Bounded run	A consecutive group of whole meetings limited by a configurable maximum meeting count.
Published results	User-visible results released after successful completion of a bounded processing run.
EARS	Easy Approach to Requirements Syntax.
SDD	Software Design Description: the implementation-facing description of components, contracts, data/class design, runtime flows and deployment.

2 Product Perspective and Context

2.1 Product Perspective

The product is a new web application that helps users follow issues across records published by the city. It complements the existing municipal record portals and provides links back to their original documents.

The selected responsibility arrangement comprises records, OCR, search, analysis, model runtime, the pipeline coordinator, the web frontend and API routing. Linking, summaries, timeline construction and meeting overviews

remain within analysis. Records owns source data and prepared content; analysis owns evidence, developments, issues and derived results; the coordinator owns durable execution state. OCR and model serving return results to the owning caller without application SQL access.

2.2 System Context



Figure 1. System context

The tracker boundary includes the application services and local inference components. Official municipal sources supply records and video references. Users inspect published results and follow links to the original sources. The figure describes information flows rather than a deployment topology.

2.3 Product Functions

The main functions are scheduled record discovery and preparation, native/OCR text extraction, chronological meeting analysis, supported development and issue identification, search across issues/records/meetings, issue histories, meeting overviews and source-grounded plain-language summaries. New or corrected sources can update the current account. A bounded run publishes its new results only after successful completion.

2.4 Stakeholders and User Classes

Table 5. Stakeholders and user classes

Stakeholder or class	Need or concern	Relationship	Source
Residents and neighborhood groups	Follow local issues and find the relevant public information.	Primary intended users	REF-04
Journalists, researchers, and students	Find related records and reconstruct the history of an issue.	Other intended users	REF-04
City staff	Retrieve information when answering public questions.	Secondary intended users	REF-04

2.5 User Characteristics

Users may be looking for information about an issue without knowing the relevant meeting dates or the terminology used in municipal records.

2.6 Operating Environment

The web frontend targets Microsoft Edge and Mozilla Firefox on user-owned devices with network access. Support for both desktop and mobile layouts remains tentative. The selected backend and persistence constraints are listed in Table 7. Each developer can use a separate local application/data instance; coordinated database-and-file snapshots support sharing prepared records and analysis results. Production hosting details remain to be specified.

3 External Interfaces

Table 6. External interfaces

ID	Interface type	Logical characteristics	Source
IF-01	User interface	A web frontend accessed through modern browsers, targeting Microsoft Edge and Mozilla Firefox. Support for both desktop and mobile layouts remains tentative.	LIM-03
IF-02	Software or API	Official records are obtained from the San Bernardino city portal. Application services exchange structured data through owner-controlled HTTP APIs. Portal endpoint details and long-running-operation contracts remain to be specified.	REF-06; LIM-09
IF-03	Hardware or device	Users access the frontend on their own device with a browser and network connection, using keyboard and mouse or touchscreen input.	REF-04
IF-04	Communications	Network communication with the San Bernardino city portal is required. Structured application inter-service payloads use HTTP/JSON; document transfer may use binary or streamed responses.	REF-06; CH-05

4 Limitations Assumptions and Dependencies

4.1 Limitations and Design Constraints

Table 7. Limitations and design constraints

ID	Limitation or constraint	Source
LIM-01	One-semester course project with two team members.	REF-01
LIM-02	The system shall use microservices and MVC architecture.	CH-02
LIM-03	The system shall provide a web frontend implemented using React, TypeScript and Vite.	CH-03
LIM-04	The system shall use Java 25 with Spring Boot 4.1.2 for the records, search, analysis and pipeline coordinator services, and Python for the OCR service.	CH-21
LIM-05	The Java services shall use Maven 3.10.0 for builds.	CH-21
LIM-06	Each local development instance shall use one PostgreSQL instance and one application database with service-owned schemas.	CH-21
LIM-07	The system shall store original document files and retained file artifacts in persistent file storage, with references and queryable metadata in SQL.	CH-21
LIM-08	The records and analysis services shall use Spring Data JPA as their primary domain-persistence approach.	CH-21
LIM-09	Other application services shall access records and analysis data through the owning services' APIs.	CH-21
LIM-10	When PDF content can be adequately recovered through native text extraction, the records service shall use native extraction instead of OCR for that content.	CH-21
LIM-11	The initial analysis design shall treat each selected meeting as a whole analysis unit.	CH-21

LIM-01 records project context. The remaining entries are selected constraints on the solution. Lower-level build organization, typed-client wiring and deployment configuration belong in the design and developer documentation.

4.2 Assumptions and Dependencies

Table 8. Assumptions and dependencies

ID	Assumption or dependency	Impact if false	Owner or action
AS-01	Enough selected City Council records can be obtained and processed by the application.	The source selection or scope may need to be narrowed.	Check that selected records can be obtained and processed.
AS-02	Issue timelines will help residents follow local matters more easily.	The proposed workflow may not meet the intended need.	Ask intended users whether issue timelines help them follow a local issue.

5 Requirements Approach and Evidence

5.1 Requirement Language and Quality Rules

Binding requirements use one shall statement and an applicable EARS pattern. Ubiquitous requirements use a direct system statement; events, states, unwanted conditions and optional features use the corresponding condition. Definitions and scope explanations remain descriptive.

Each requirement is reviewed for necessity, clarity, singularity, correctness, consistency, traceability, feasibility and verifiability. Requirement-set review checks coverage and conflicting obligations. Course-tailored ISO/IEC/IEEE 29148:2018 guidance is used without claiming formal certification.

5.2 Evidence Register

Table 9. Evidence register

ID	Evidence record	Date and owner	Contribution	Status
EV-01	Project Pitch and Speaker Notes identified by REF-04	Sep. - Kazim Yigit Ablay	Establishes the proposed purpose and initial features	Available and Reviewed
EV-02	Supplied CSE 6550 SRS Template	Oct. 5 - Both	Required document structure and requirement quality guidance	Available and Reviewed
EV-03	Inspection of selected official records in the PR-01 feasibility investigation (REF-09)	October 11, 2026; Kazim Yigit Ablay	Demonstrates selected downloads, native/HTML extraction, retained source relationships and local OCR experiments. Does not establish broad extraction completeness or official transcript acquisition.	Available; selected examples reviewed
EV-04	Feedback from Intended City Users	TBD - Michael Avila	Evaluates whether the issue timelines help users answer constituent questions	Planned

5.3 Requirement Attributes

Current requirement version is 1.0 and status is Draft for team review. Requirements owners remain Kazim Yigit Ablay and Michael Avila. Conditional categorization is identified in the traceability matrix. Existing Must priorities and High risk values are retained; new priority and risk values are TBD until agreed. TBD means to be determined, not a priority level or an acceptance result.

Source/basis entries identify an actual reference, parent requirement, scope section or recorded baseline change in Appendix A. Recorded baseline choices establish design intent; they are distinct from empirical stakeholder evidence. Verification methods are T (test), I (inspection), A (analysis) and D (demonstration). All verification remains planned unless a result is explicitly recorded.

6 Specified Requirements

6.1 Functional Requirements

Table 10. Functional requirements

ID	Requirement	Source	Priority	Risk	Verify
FR-01	The system shall import selected official agendas, minutes, staff reports, attachments, existing official transcripts and meeting-cancellation notices.	REF-04; CH-06	Must	TBD	T
FR-02	The system shall allow users to search for issues, individual records and meetings.	REF-04; CH-07	Must	TBD	T
FR-03	The system shall identify source-supported connections among records concerning the same issue.	REF-04; CH-08	Must	High	T
FR-04	When displaying an issue timeline, the system shall present the issue's supported evolution across the imported records.	REF-04; CH-09	Must	TBD	D
FR-05	The system shall provide meeting overviews covering agenda topics and recorded outcomes.	§1.3; CH-10	Must	TBD	T
FR-06	The system shall provide plain-language summaries supported by the official records.	REF-04; CH-11	Must	High	T
FR-07	When presenting source-supported information, the system shall provide links to the original official records supporting that information.	REF-04; CH-12	Must	TBD	T
FR-08	When an official meeting-video link is available, the system shall provide that link.	§1.3; CH-13	Must	TBD	T
FR-09	When reconstructing an issue's history, the system shall use supporting official records currently available to it, including later records that establish earlier developments.	FR-04; CH-21	TBD	TBD	T
FR-10	When a scheduled discovery run begins, the records service shall inspect the selected official sources for new or updated in-scope records.	FR-01; CH-21	TBD	TBD	T
FR-11	When a new expected in-scope record is discovered, the records service shall retrieve it for preparation.	FR-01; CH-21	TBD	TBD	T
FR-12	When selected source content requires OCR, the records service shall request its text extraction from the OCR service.	FR-01; CH-21	TBD	TBD	T
FR-13	While the active run's selected inputs remain unprepared, the system shall keep meeting analysis pending.	FR-01; CH-21	TBD	TBD	T
FR-14	When processing an ordered meeting collection, the system shall use official meeting date as the primary analysis-order key.	FR-01; CH-21	TBD	TBD	T
FR-15	While the current meeting's analysis or saved result updates remain unfinished, the coordinator shall keep analysis of the next meeting pending.	FR-01; CH-21	TBD	TBD	T

ID	Requirement	Source	Priority	Risk	Verify
FR-16	When forming processing runs, the coordinator shall group consecutive whole meetings up to a configurable maximum meeting count.	FR-01; CH-21	TBD	TBD	T
FR-17	When discovery finds no new or changed records and no pending work requires continuation, the system shall not start fresh analysis for unchanged inputs.	FR-01; CH-21	TBD	TBD	T
FR-18	When a record type has not been published for a selected meeting, the system shall process the available expected inputs without requiring a complete agenda/minutes/transcript bundle.	FR-01; CH-21	TBD	TBD	T
FR-19	When official evidence establishes cancellation, the system shall represent the scheduled meeting as cancelled.	FR-01; CH-21	TBD	TBD	T
FR-20	When a cancelled meeting's applicable expected inputs have been handled successfully, the coordinator shall allow progression beyond that scheduled meeting.	FR-01; CH-21	TBD	TBD	T
FR-21	When the system establishes a source-supported inferred issue association, the system shall include that association in the issue timeline.	FR-03; CH-21	TBD	TBD	T
FR-22	When presenting a connection, the system shall distinguish a source-stated relationship from a system-inferred relationship.	FR-03; CH-21	TBD	TBD	D
FR-23	When presenting a connection, the system shall provide access to its supporting source passages.	FR-03; CH-21	TBD	TBD	T
FR-24	If successfully processed sources provide insufficient support for an issue association, then the system shall leave that association unset.	FR-03; CH-21	TBD	TBD	T
FR-25	When new or corrected official records become available, the system shall update affected issue histories, timelines, meeting overviews and summaries where the evidence warrants a change.	§1.3; CH-21	TBD	TBD	T
FR-26	When a newer official source explicitly corrects an earlier claim, the system shall reflect the correction in the current account.	§1.3; CH-21	TBD	TBD	T
FR-27	When comparable official claims conflict without an explicit correction, the system shall prefer the newer publication as the latest reported account.	§1.3; CH-21	TBD	TBD	T
FR-28	When presenting an unresolved contradiction, the system shall identify the disagreement with access to both conflicting sources.	§1.3; CH-21	TBD	TBD	D
FR-29	When official evidence establishes a development's occurrence date, the system shall use that date in the issue timeline.	FR-04; CH-21	TBD	TBD	T
FR-30	If a development's occurrence date is unavailable, then the system shall display the source meeting date as a reported or discussed date.	FR-04; CH-21	TBD	TBD	T

ID	Requirement	Source	Priority	Risk	Verify
FR-31	When a development has an approximate date or date range, the system shall display it at the precision supplied by the source.	FR-04; CH-21	TBD	TBD	T
FR-32	When displaying a source-supported future event date, the system shall identify the event as planned or expected.	FR-04; CH-21	TBD	TBD	T
FR-33	When a bounded run completes successfully, the system shall make that run's new results available to users.	§1.3; CH-21	TBD	TBD	T
FR-34	While a meeting-processing failure remains unresolved, the coordinator shall keep later meetings and runs pending.	§1.3; CH-21	TBD	TBD	T
FR-35	When analysis succeeds with explicitly unknown findings or an unsupported association left unset, the coordinator shall permit progression.	§1.3; CH-21	TBD	TBD	T
FR-36	When a user selects one or more search types, the system shall restrict returned results to the selected types.	FR-02; CH-21	TBD	TBD	T
FR-37	While an imported record is published, the system shall keep it accessible regardless of whether it has an issue association.	FR-02; CH-21	TBD	TBD	T
FR-38	When selected sources do not establish an outcome, the system shall describe the outcome as unavailable in those sources.	FR-05; CH-21	TBD	TBD	T
FR-39	Where issue categorization is included, the system shall assign issue categories from the agreed flat category list.	§1.3; CH-21	TBD	TBD	T
FR-40	Where issue categorization is included, if category evidence is unclear, then the system shall leave the issue uncategorized.	§1.3; CH-21	TBD	TBD	T
FR-41	When analyzing selected records, the analysis service shall extract source-supported developments separately from their supporting source passages.	FR-03; CH-21	TBD	TBD	T/I
FR-42	The records service shall acknowledge preparation completion only after validated prepared content has been durably saved.	FR-01; CH-21	TBD	TBD	T

FR-39 and FR-40 apply where categorization is included under the agreed simplicity condition. Search result types and their combinations are specified; detailed matching, ranking and result presentation remain for Michael-led search design.

6.2 Data Requirements

Table 11. Data requirements

ID	Requirement	Source	Priority	Verify
DR-01	The system shall retain imported records with information identifying their original sources.	REF-04; FR-07	Must	T/I

ID	Requirement	Source	Priority	Verify
DR-02	The system shall retain imported original document bytes in persistent storage.	DR-01; CH-21	TBD	T/I
DR-03	The system shall retain SQL references and identifying metadata for stored source files and retained artifacts.	DR-01; CH-21	TBD	I
DR-04	The system shall identify source meetings using the official portal meeting identifier within its source-portal context.	DR-01; CH-21	TBD	T
DR-05	The system shall preserve official relationships among meetings, documents, agenda items and attachments.	DR-01; CH-21	TBD	T
DR-06	The records service shall retain prepared readable content with its original-record association.	DR-01; CH-21	TBD	I
DR-07	The system shall represent source evidence, developments and ongoing issues as distinct linked concepts.	DR-01; CH-21	TBD	I
DR-08	The system shall retain an identifiable supporting source passage for each evidence item.	DR-01; CH-21	TBD	T/I
DR-09	The system shall distinguish development dates, source meeting dates and available source publication dates in its retained data.	DR-01; CH-21	TBD	T
DR-10	The system shall preserve the precision, range or unknown state of each retained development date.	DR-01; CH-21	TBD	T
DR-11	When an official source is corrected or replaced, the system shall retain the earlier imported source version.	DR-01; CH-21	TBD	T
DR-12	The system shall retain the source-version associations supporting its current issue histories and derived results.	DR-01; CH-21	TBD	T/I
DR-13	The pipeline coordinator shall persist the execution and progression information needed for run recovery.	DR-01; CH-21	TBD	T
DR-14	Where issue categorization is included, the system shall restrict category values to the agreed initial vocabulary.	DR-01; CH-21	TBD	I
DR-15	Where issue categorization is included, the system shall support zero, one or several categories per issue without a hierarchy or required primary category.	DR-01; CH-21	TBD	T
DR-16	The system shall retain a cancelled meeting's identity and official cancellation evidence.	DR-01; CH-21	TBD	I
DR-17	A shared development dataset shall contain a database snapshot and the matching retained files referenced by that snapshot.	DR-01; CH-21	TBD	T

For conditional DR-14/DR-15 and FR-39, the flat category vocabulary is Transportation and roads; Housing and homelessness; Planning and land use; Public safety; Parks and community services; Utilities and environment; Economic development; Budget and finance; City government. No primary category or hierarchy is required.

6.3 Nonfunctional and Quality Requirements

Table 12. Quality requirements

ID	Quality	Requirement	Source	Verify
NFR-01	Factual / temporal integrity	When generating summaries or meeting overviews, the system shall base factual claims on the selected official records.	REF-04; CH-15	T/I
NFR-02	Security	When displaying imported document text, the system shall prevent embedded scripts from executing.	MU-05; CH-16	T/I
NFR-03	Accessibility	The system shall allow users to complete search, record inspection, and source-link navigation using only a keyboard.	CH-17	T/I
NFR-04	Reliability	If import or meeting processing fails, then the system shall preserve previously saved records, source references and completed work for recovery.	FR-01; CH-18	T/I
NFR-05	Factual / temporal integrity	When generating summaries or meeting overviews, the system shall preserve material qualifications that affect the meaning of the source information.	NFR-01; CH-21	I
NFR-06	Factual / temporal integrity	When describing official-record content, the system shall distinguish proposals, recommendations and plans from documented actions.	NFR-01; CH-21	I
NFR-07	Factual / temporal integrity	When a planned date passes, the system shall preserve the event's unconfirmed occurrence status unless later official evidence establishes a change.	NFR-01; CH-21	T
NFR-08	Factual / temporal integrity	When coarse or overlapping dates do not establish an exact sequence, the timeline shall preserve that ordering uncertainty.	NFR-01; CH-21	I
NFR-09	Factual / temporal integrity	If a meeting is cancelled, then the system shall not infer cancellation of its underlying issues solely from that meeting cancellation.	NFR-01; CH-21	T
NFR-10	Reliability	While a new run is processing or awaiting recovery, the system shall keep the last successfully published results available for browsing and search.	NFR-04; CH-21	T
NFR-11	Reliability	While a bounded run is unfinished or unsuccessful, the system shall keep its incomplete new results unavailable to users.	NFR-04; CH-21	T
NFR-12	Reliability	While a processing run is active or its failure remains unresolved, the coordinator shall prevent another processing run from starting.	NFR-04; CH-21	T
NFR-13	Reliability	While processing is active, the system shall serve browsing and search requests from published results without waiting for the entire processing run to finish.	NFR-04; CH-21	T
NFR-14	Security	When preparing model input, the analysis service shall separate trusted processing instructions from imported record content.	NFR-02; CH-21	T/I

ID	Quality	Requirement	Source	Verify
NFR-15	Security	If imported content contains instructions, then the system shall not grant those instructions authority to execute commands, modify application data or bypass validation.	NFR-02; CH-21	T
NFR-16	Security	When OCR or model inference returns output, the owning service shall validate it before accepting it as a usable persisted result.	NFR-02; CH-21	T
NFR-17	Security	The OCR service and model runtime shall have no application SQL access.	NFR-02; CH-21	T/I

State-based reliability and security checks have observable pass/fail conditions. Semantic source-support and material-qualification review still require a documented rubric and reviewed reference examples. Aggregate acceptance targets are agreed using development evaluation before final acceptance; no numerical model accuracy, extraction quality, uptime or response-time target is asserted here. Schema-valid output alone is not evidence of factual correctness.

6.4 Standards and Policy Requirements

Table 13. Standards and policy requirements

ID	Requirement	Source
COM-01	The team shall maintain this SRS and its requirement records using the course-tailored practices derived from ISO/IEC/IEEE 29148:2018.	REF-01; REF-03
COM-02	The team shall document significant generative AI assistance, human review, validation actions, and responsibility in the AI Assistance and Provenance record.	REF-01
COM-03	Project Java source shall use the Google Java Style Guide as its coding-style basis.	REF-07; CH-21
COM-04	Project Java packages shall use org.civictracker as their base namespace.	CH-21
COM-05	The team shall maintain selected project engineering documents in the version-controlled docs/engineering directory.	CH-21
COM-06	The team shall express each binding requirement as a single statement using an applicable EARS pattern.	REF-02; REF-10
COM-07	When revising an existing requirement, the team shall preserve its stable identifier.	REF-02; REF-10
COM-08	When the SRS baseline changes, the team shall record the affected requirements and reasons in its revision history.	REF-02; REF-10
COM-09	The team shall maintain traceability from active requirements to their source basis, design allocation and planned verification.	REF-02; REF-10
COM-10	The team shall use reviewed official-record examples as the reference basis for linking evaluation.	CH-21
COM-11	For each linking evaluation, the team shall record its inputs, reference judgments and model/prompt configuration.	CH-21
COM-12	For linking evaluation, the team shall report precision, recall, evidence presence and evidence support for the defined association scope.	CH-21
COM-13	The team shall exclude machine-specific local file locations from authored engineering documents, source code and configuration files where possible.	CH-21

7 User Stories and Acceptance Criteria

User stories express stakeholder value. The listed acceptance scenarios provide observable examples and reference the controlling requirements. They are planned checks rather than completed user validation.

Table 14. User stories

ID	User story	Acceptance criteria	Traces to
US-01	As a resident, I want to find records about a local issue so that I can begin investigating without knowing its meeting date.	1. Given published records about a known local issue, when the user searches without entering a meeting date, then relevant records and their original source links can be opened. 2. Given matching issues, records and meetings in the published test dataset, when the user selects one or more search types, then returned results contain only selected types. 3. Given a published record without an issue association, when the user opens it through record search, then its original source remains accessible.	FR-02, FR-07, FR-36, FR-37
US-02	As a resident, I want to see why two records are connected and whether the connection is uncertain so that I can judge how much to rely on it.	1. Given a source-stated connection and an inferred connection, when each is presented, then its relationship basis is distinguishable and its supporting source passages are accessible. 2. Given successfully analyzed sources that do not sufficiently support an issue association, then the association remains unset rather than appearing as an established timeline connection.	FR-03, FR-21, FR-22, FR-23, FR-24
US-03	As a resident, I want to view an issue timeline so that I can follow the issue across meetings.	1. Given records describing a supported issue development, when its timeline is opened, then the development appears as part of the issue's evolution with access to its original sources. 2. Given a source-established occurrence date, then the timeline uses that date; where it is unavailable, the source meeting date is labeled reported or discussed. 3. Given approximate/ranged dates, then the timeline preserves their precision and does not assert an unsupported exact sequence. 4. Given a source-supported future plan, then it is identified as planned or expected; advancing time alone does not establish completion, cancellation or failure.	FR-04, FR-07, FR-29, FR-30, FR-31, FR-32, NFR-07, NFR-08

8 Misuse and Abuse Cases

Table 15. Misuse and abuse cases

ID	Scenario	Affected asset	Required treatment	Traces to
MU-01	A meeting overview presents a proposed agenda item as approved action when no selected record documents approval.	Accuracy of meeting outcomes; user understanding	Distinguish agenda topics from documented outcomes. Indicate when outcomes are unavailable in the selected records.	FR-05, FR-38, NFR-01, NFR-06
MU-02	Records about different projects have similar names and are incorrectly connected, causing a user to misunderstand an issue's history.	Accuracy of issue timelines	Distinguish source-stated and system-inferred connections and provide their supporting passages. Leave unsupported associations unset.	FR-03, FR-22, FR-23, FR-24
MU-03	A summary adds an unsupported detail or omits a qualification that changes the meaning of the source.	Summary Accuracy; trust in the application	Provide supporting source references and evaluate summaries against their records, including important qualifications.	FR-06, NFR-01, NFR-05, NFR-06
MU-04	A user assumes that no search results means the city never discussed an issue, although the relevant records were not imported.	User understanding of coverage	Explain that results cover selected imported records and make dataset coverage visible.	FR-01, FR-02, DR-01
MU-05	An imported document contains script markup that executes when its text is displayed.	Browser security; application integrity	Display imported content without executing embedded scripts.	NFR-02
MU-06	Instructions in an imported record attempt to override analysis instructions, authorize application actions or bypass validation.	Application authority; analysis integrity	Treat source text as untrusted data; enforce authority and output validation in application boundaries. Prompt separation alone is not a security guarantee.	NFR-14, NFR-15, NFR-16, NFR-17

9 Prototype and Requirements Validation Plan

Table 16. Prototype and validation plan

ID	Uncertainty	Prototype	Evaluator and task	Decision evidence	Traces to
PR-01	Can selected official records be obtained and processed with enough source information?	Small import experiment using representative records	The project team imports selected documents and compares their contents and source information with the originals.	Selected-record feasibility investigation completed (EV-03/REF-09). Native/HTML extraction and OCR examples were examined; broader completeness and transcript acquisition remain to verify.	FR-01, DR-01, FR-07, LIM-10

ID	Uncertainty	Prototype	Evaluator and task	Decision evidence	Traces to
PR-02	Can users find an issue without knowing its meeting dates or official terminology?	Search prototype using a small record collection	Intended users search for a known issue using their own words and identify relevant records.	Michael-led search evaluation. Record successful/unsuccessful queries, missed records and terminology problems; participant outcomes remain to be documented before claiming validation.	FR-02
PR-03	Can users understand why records are connected and recognize that a connection is tentative?	Mockup showing suggested connections and supporting evidence	Intended users inspect a plausible connection and a deliberately questionable connection, then explain whether they would rely on each.	Record whether users understand the source evidence and distinguish source-stated from system-inferred relationships. Revise presentation if they do not.	FR-03, FR-22, FR-23, FR-24
PR-04	Does the timeline help users follow an issue across meetings, and which date is most understandable?	Timeline mockup containing records from several meetings	Intended users describe the issue's sequence of events and explain what the displayed dates mean.	Check users' understanding of issue evolution, supported dates, reported/discussed fallback and approximate/planned entries. Behavioral date policy is resolved; usability validation remains planned.	FR-04, FR-29–FR-32, NFR-07, NFR-08
PR-05	Can users distinguish agenda topics from documented outcomes in an overview and check its summary?	Meeting overview mockup with summaries and source links, including an item without a documented outcome	Intended users identify what was proposed, what outcome is documented, and where they can verify the information. The team also compares statements with the sources.	Record unsupported statements, mistaken interpretations, and difficulty locating evidence. Revise overview wording, missing-outcome handling, or source presentation.	FR-05, FR-06, FR-07, NFR-01

10 Requirement Allocation and Release Planning

Table 17. Preliminary allocation

Allocation target	Requirements	Release	Rationale
Records	FR-01, FR-10–FR-12, FR-18, FR-19, FR-42; DR-01–DR-06, DR-11, DR-16	Not allocated	Own official-source identities, files, prepared content and durable completion.
Analysis	FR-03–FR-06, FR-09, FR-21–FR-32, FR-38–FR-41; DR-07–DR-10, DR-12, DR-14, DR-15	Not allocated	Keep linking, summaries, timelines and overviews together; categorization remains conditional within existing analysis.

Allocation target	Requirements	Release	Rationale
Search	FR-02, FR-36, FR-37; LIM-09	Not allocated	Obtain owner data through APIs; Michael leads detailed matching/ranking design.
Pipeline coordinator	FR-13–FR-17, FR-20, FR-33–FR-35; DR-13; NFR-04, NFR-10–NFR-13	Not allocated	Own preparation/analysis progression, run admission and recovery coordination.
OCR and model runtime	LIM-04; NFR-17	Not allocated	Return inference results to records/analysis; no application SQL access.
Frontend and API routing	FR-07, FR-08, FR-22, FR-23, FR-29–FR-32, FR-36, FR-37; NFR-02, NFR-03	Not allocated	Present published results, evidence and selectable search types.
Project process	COM-01–COM-13; DR-17	Not allocated	Maintain engineering records, evaluation references and coordinated development datasets.

Allocation identifies component responsibilities, not human implementation assignments or release commitments. Release scheduling remains to be agreed. The selected Java build uses a root parent/aggregator and a child module per Java service; typed Spring HTTP interfaces backed by RestClient are the ordinary Java client approach. Detailed contracts and build/deployment configuration belong in the SDD and developer guidance.

11 Verification and Life Cycle Traceability

11.1 Verification Methods

Table 18. Verification methods

Method	Meaning
Test	Use controlled inputs and compare the observed behavior with the expected result.
Inspection	Examine an interface, record, document, or implementation for the required information.
Analysis	Evaluate recorded data or measurements against the requirement.
Demonstration	Show the required capability in use.

Linking evaluation uses reviewed official-record references with an explicit association unit, expected associations and ambiguous judgments. Development examples remain separate from final evaluation examples. Numerical acceptance targets are agreed using development results before the final acceptance run. Inputs, judgments and model/prompt configuration are recorded. Precision, recall, evidence presence and evidence support are reported with their denominators. These are planned evaluation procedures; no results or numeric thresholds are invented.

For the defined association scope, precision is correct returned associations divided by returned associations; recall is correct returned associations divided by expected reference associations. Evidence presence measures returned associations with usable references. Evidence support measures reviewed associations whose cited passages justify the connection, divided by all reviewed returned associations. Missing evidence does not count as supported. Report ambiguous/excluded cases and undefined zero-denominator measures explicitly.

Cross-run exclusion, publication isolation and recovery require application-level tests. Spring Batch remains conditional on that proof; framework protection of the same job instance does not establish exclusion across different runs. Whole-meeting model capacity, adequate native/OCR extraction and usable model-output validation remain to be demonstrated with representative inputs.

V-16 Discovery and input preparation. Check expected-input discovery, native/OCR preparation, unpublished-source handling and acknowledgment only after durable saves. Status: planned.

V-17 Chronological execution and recovery. Check meeting-date order, consecutive bounded groups, full meeting completion, durable progress and cross-run exclusion, including unresolved remote work. Status: planned.

V-18 Cancelled meeting handling. Check official cancellation evidence, retained meeting identity, no invented agenda outcomes and safe progression. Status: planned.

V-19 Connection evidence and abstention. Check explicit/inferred presentation, passage access, development/evidence distinction and unset insufficient-support associations. Status: planned.

V-20 Later sources and contradictions. Check retrospective evidence, source-version retention, explicit corrections, newer-publication preference and unresolved disagreement display. Status: planned.

V-21 Timeline date semantics. Check occurrence versus source dates, labeled fallback, precision/ranges, uncertain ordering and planned dates without time-triggered completion. Status: planned.

V-22 Publication and failure visibility. Check full-run publication, prior-view availability, hidden incomplete results, failure blocking and nonblocking valid uncertainty. Status: planned.

V-23 Selectable search types. Check issue/record/meeting selection combinations and published unassociated-record access without selecting Michael's matching/ranking implementation. Status: planned.

V-24 Retained data and provenance. Inspect each data requirement with its per-item checks; this umbrella plan is not a claim that one test proves all data properties. Status: planned.

V-25 Conditional categorization. If included, check accepted labels, flat multiple membership, unclear cases and assignment within existing analysis. Status: planned.

V-26 Model and service trust boundaries. Check source/instruction separation, authority enforcement, output validation and database-free inference. Status: planned.

V-27 Selected constraints and conventions. Inspect selected technologies/ownership/interfaces without treating a configuration inspection as proof of model quality or runtime feasibility. Status: planned.

V-28 Development dataset restoration. Restore a coordinated database/files snapshot, inspect saved source/results references and avoid automatic resumption of copied control state. Status: planned.

11.2 Traceability Matrix

Each core entry is mapped below to its basis, allocated responsibility and planned verification. V-01 through V-15 retain the submitted baseline meanings; V-16 through V-28 identify the additional planned scenarios above. A planned reference is not an executed test or a passing result.

Table 19. Traceability matrix

Source / basis	Req. ID	Design / allocation	Planned verification	Status
REF-01	LIM-01	Project process	V-14; Inspect course/team context.	Planned
CH-02	LIM-02	Architecture	V-27; Inspect architecture selection.	Planned
CH-03	LIM-03	Architecture	V-27; Inspect web frontend selection.	Planned
CH-21	LIM-04	Architecture	V-27; each Java service build declares the selected baseline. Inspection: OCR service implementation and dependency configuration use Python.	Planned
CH-21	LIM-05	Architecture	V-27; documented build invocation and build-tool version match the selection.	Planned
CH-21	LIM-06	Architecture	V-27, V-28; development database layout matches the selected topology.	Planned
CH-21	LIM-07	Architecture	V-27; file data and SQL references have the selected separation.	Planned
CH-21	LIM-08	Architecture	V-27; ordinary persistence uses service-local JPA entities and repositories.	Planned

Source / basis	Req. ID	Design / allocation	Planned verification	Status
CH-21	LIM-09	Architecture	V-26, V-27; analysis reads use records-owned endpoints. Inspection/test: search integration reads use owner APIs.	Planned
CH-21	LIM-10	Architecture	V-16, V-27; native-text and scanned examples exercise native-first preparation.	Planned
CH-21	LIM-11	Architecture	V-17, V-27; Inspection/analysis: initial processing scope is the full selected meeting.	Planned
REF-04; CH-06	FR-01	Records / OCR	V-01, V-16; Import selected records and inspect source retention.	Planned
REF-04; CH-07	FR-02	Search / frontend	V-02, V-23; Search agreed reference queries and inspect returned types.	Planned
REF-04; CH-08	FR-03	Analysis / frontend	V-03, V-19; Compare links/evidence with reviewed reference relationships.	Planned
REF-04; CH-09	FR-04	Analysis / frontend	V-04, V-21; Show issue evolution and correct date meaning.	Planned
§1.3; CH-10	FR-05	Analysis / frontend	V-05; Compare agenda topics/outcomes with official records.	Planned
REF-04; CH-11	FR-06	Analysis / frontend	V-06; Review plain language and source-supported meaning.	Planned
REF-04; CH-12	FR-07	Analysis / frontend	V-07; Check original-source targets.	Planned
§1.3; CH-13	FR-08	Analysis / frontend	V-08; Check known official video targets.	Planned
FR-04; CH-21	FR-09	Analysis / frontend	V-20; a later source can support an earlier development without changing its source attribution.	Planned
FR-01; CH-21	FR-10	Records / OCR	V-16; known and newly published records are distinguished during discovery.	Planned
FR-01; CH-21	FR-11	Records / OCR	V-16; a discovered expected record enters retrieval and preparation.	Planned
FR-01; CH-21	FR-12	Records / OCR	V-16; scanned content invokes OCR and returns to records for handling.	Planned
FR-01; CH-21	FR-13	Coordinator	V-16; no meeting analysis starts while required run-input preparation remains unfinished.	Planned
FR-01; CH-21	FR-14	Coordinator	V-17; meetings with distinct known dates are analyzed oldest to newest.	Planned
FR-01; CH-21	FR-15	Coordinator	V-17; next-meeting inference waits for current meeting analysis and owner saves.	Planned
FR-01; CH-21	FR-16	Coordinator	V-17; ordered inputs are partitioned without splitting meetings or exceeding the configured count.	Planned
FR-01; CH-21	FR-17	Coordinator	V-16; an unchanged, fully completed dataset starts no fresh analysis; a pending recovery/update is not suppressed.	Planned

Source / basis	Req. ID	Design / allocation	Planned verification	Status
FR-01; CH-21	FR-18	Records / OCR	V-16; an agenda-only source set can complete with outcome information unavailable.	Planned
FR-01; CH-21	FR-19	Records / OCR	V-18; cancellation evidence produces a cancelled meeting representation.	Planned
FR-01; CH-21	FR-20	Coordinator	V-18; a properly accounted-for cancellation does not block later meetings.	Planned
FR-03; CH-21	FR-21	Analysis / frontend	V-19; a supported inferred association appears in the timeline.	Planned
FR-03; CH-21	FR-22	Analysis / frontend	V-19; a user can identify the relationship basis for explicit and inferred examples.	Planned
FR-03; CH-21	FR-23	Analysis / frontend	V-19; both source-stated and inferred examples lead to their supporting passages.	Planned
FR-03; CH-21	FR-24	Analysis / frontend	V-19; an inconclusive reference case is not forced into an issue timeline.	Planned
§1.3; CH-21	FR-25	Analysis / frontend	V-20; a later/corrected source changes the affected account while retaining source provenance.	Planned
§1.3; CH-21	FR-26	Analysis / frontend	V-20; an explicit correction replaces the superseded current claim.	Planned
§1.3; CH-21	FR-27	Analysis / frontend	V-20; an unexplained same-fact conflict uses the later publication as the latest account.	Planned
§1.3; CH-21	FR-28	Analysis / frontend	V-20; conflicting comparable claims and both sources remain inspectable.	Planned
FR-04; CH-21	FR-29	Analysis / frontend	V-21; a retrospectively reported development uses its supported occurrence date.	Planned
FR-04; CH-21	FR-30	Analysis / frontend	V-21; an unknown-date example has a labeled fallback rather than a claimed occurrence date.	Planned
FR-04; CH-21	FR-31	Analysis / frontend	V-21; month/year and interval examples display without fabricated exact days.	Planned
FR-04; CH-21	FR-32	Analysis / frontend	V-21; a proposed future start is visibly distinguished from a documented occurrence.	Planned
§1.3; CH-21	FR-33	Coordinator	V-22; complete run results become visible together; NFR-10/NFR-11 govern prior state and failures.	Planned
§1.3; CH-21	FR-34	Coordinator	V-22; failed expected-input preparation blocks later progression until resolved.	Planned
§1.3; CH-21	FR-35	Coordinator	V-22; genuine uncertainty after successful analysis does not become a processing failure.	Planned
FR-02; CH-21	FR-36	Search / frontend	V-23; each single type, each combination and all three respect the selected scope.	Planned
FR-02; CH-21	FR-37	Search / frontend	V-23; an unassociated published record remains discoverable and openable.	Planned
FR-05; CH-21	FR-38	Analysis / frontend	V-05, V-10; absent minutes do not produce a claim that no action occurred.	Planned

Source / basis	Req. ID	Design / allocation	Planned verification	Status
§1.3; CH-21	FR-39	Analysis / frontend	V-25; assigned category values belong to the agreed vocabulary.	Conditional; planned
§1.3; CH-21	FR-40	Analysis / frontend	V-25; an unclear case receives no forced category.	Conditional; planned
FR-03; CH-21	FR-41	Analysis / frontend	V-19; a reviewed proposal/decision/change is represented as a development linked to evidence.	Planned
FR-01; CH-21	FR-42	Records / OCR	V-16; a failed owner save cannot produce a successful preparation-completion acknowledgment.	Planned
REF-04; FR-07	DR-01	Records	V-09, V-24; Inspect retained source-identifying information.	Planned
DR-01; CH-21	DR-02	Records	V-24; retained originals match the downloaded source bytes.	Planned
DR-01; CH-21	DR-03	Records	V-24; SQL records identify the retained files and their source associations.	Planned
DR-01; CH-21	DR-04	Records	V-24; distinct official IDs remain distinct even when meeting dates coincide.	Planned
DR-01; CH-21	DR-05	Records	V-24; a known attachment retains its actual official meeting/item relationship.	Planned
DR-01; CH-21	DR-06	Records	V-24; saved extraction output resolves back to its source record.	Planned
DR-01; CH-21	DR-07	Analysis	V-19, V-24; the domain representation can distinguish a source passage, an occurrence/proposal and its ongoing matter.	Planned
DR-01; CH-21	DR-08	Analysis	V-19, V-24; evidence resolves to an identifiable passage in its actual source.	Planned
DR-01; CH-21	DR-09	Analysis	V-21, V-24; a later-published report about an earlier event retains distinct date roles.	Planned
DR-01; CH-21	DR-10	Analysis	V-21, V-24; month-only, year-only, interval and unknown values survive save/load.	Planned
DR-01; CH-21	DR-11	Records	V-20, V-24; an update retains the prior source version.	Planned
DR-01; CH-21	DR-12	Analysis	V-20, V-24; current and corrected claims can be traced to their actual supporting versions.	Planned
DR-01; CH-21	DR-13	Coordinator	V-17, V-24; persisted run progress remains available after coordinator restart.	Planned
DR-01; CH-21	DR-14	Analysis	V-24, V-25; allowed values match the nine labels in the category dictionary.	Conditional; planned
DR-01; CH-21	DR-15	Analysis	V-24, V-25; zero, one and several categories can be represented without a primary-category obligation.	Conditional; planned
DR-01; CH-21	DR-16	Records	V-18, V-24; cancellation remains associated with the correct scheduled meeting and source.	Planned

Source / basis	Req. ID	Design / allocation	Planned verification	Status
DR-01; CH-21	DR-17	Development	V-24, V-28; referenced originals/artifacts and saved processed results are available after import.	Planned
REF-04; CH-15	NFR-01	Analysis	V-10; Review factual claims and preserved qualifications.	Planned
MU-05; CH-16	NFR-02	Frontend	V-11; Display script-bearing text without script execution.	Planned
CH-17	NFR-03	Frontend	V-12; Complete search/record/source navigation by keyboard.	Planned
FR-01; CH-18	NFR-04	Coordinator	V-13, V-17, V-22; Check saved records and completed work after failure.	Planned
NFR-01; CH-21	NFR-05	Analysis	V-10; reviewed source/output pairs retain meaning-changing qualifications.	Planned
NFR-01; CH-21	NFR-06	Analysis	V-10; proposal/decision reference cases retain the correct factual status.	Planned
NFR-01; CH-21	NFR-07	Analysis	V-21; advancing the clock alone does not create completion, cancellation or failure.	Planned
NFR-01; CH-21	NFR-08	Analysis	V-21; ambiguous date examples do not assert unsupported before/after relationships.	Planned
NFR-01; CH-21	NFR-09	Analysis	V-18; meeting cancellation alone does not change a project's substantive outcome.	Planned
NFR-04; CH-21	NFR-10	Coordinator	V-22; prior published results remain accessible during a controlled unfinished/failed run.	Planned
NFR-04; CH-21	NFR-11	Coordinator	V-22; public reads expose none of the unfinished run's new results.	Planned
NFR-04; CH-21	NFR-12	Coordinator	V-17; overlapping requests and launch paths do not start a second run.	Planned
NFR-04; CH-21	NFR-13	Coordinator	V-22; an interactive request can finish before a deliberately unfinished processing run.	Planned
NFR-02; CH-21	NFR-14	Trust boundary	V-26; source text occupies an explicitly untrusted data boundary.	Planned
NFR-02; CH-21	NFR-15	Trust boundary	V-26; controlled injected instructions cannot authorize those application actions.	Planned
NFR-02; CH-21	NFR-16	Trust boundary	V-26; malformed/unusable controlled responses are not saved as successful domain results.	Planned
NFR-02; CH-21	NFR-17	Trust boundary	V-26; inference components cannot read or change application SQL data.	Planned
REF-01; REF-03	COM-01	Project process	V-14; Inspect course-tailored requirements records.	Planned
REF-01	COM-02	Project process	V-15; Inspect current and historical AI-provenance records.	Planned
REF-07; CH-21	COM-03	Project process	V-27; Java naming/formatting follows the adopted guide.	Planned

Source / basis	Req. ID	Design / allocation	Planned verification	Status
CH-21	COM-04	Project process	V-27; service packages use the selected base namespace.	Planned
CH-21	COM-05	Project process	V-14; maintained SRS/SPMP/ADR artifacts belong in the agreed engineering set.	Planned
REF-02; REF-10	COM-06	Project process	V-14; each binding statement has one shall and an appropriate EARS pattern.	Planned
REF-02; REF-10	COM-07	Project process	V-14; revised existing requirements keep IDs; genuine splits/replacements get documented mappings.	Planned
REF-02; REF-10	COM-08	Project process	V-14; revision history explains the actual v0.2-to-v1.0 changes.	Planned
REF-02; REF-10	COM-09	Project process	V-14; each active requirement has valid backward/forward references.	Planned
CH-21	COM-10	Project process	V-14; reference examples are reviewed official-record cases; development and final examples remain distinct.	Planned
CH-21	COM-11	Project process	V-14; the evaluation record identifies the compared dataset/reference/configuration.	Planned
CH-21	COM-12	Project process	V-14; recompute all four measures from returned and expected reference associations.	Planned
CH-21	COM-13	Project process	V-14; authored documents contain portable references rather than machine-specific locations.	Planned

12 Supporting Information and Open Questions

12.1 Supporting Information

Supporting material comprises the project pitch and speaker notes, the submitted v0.2 SRS, the PR-01 team review handoff and selected official source records. Interface contracts, reviewed evaluation reference sets and execution test reports form part of the engineering dossier as they are produced.

12.2 Open Questions

Table 20. Open questions and resolutions

ID	Question	Owner	Decision point	Affected area
OQ-01	How will we decide when records can be linked automatically? Partially resolved. Inferred links, visible relationship basis, evidence access and abstention are decided. The evidence rubric, reference association unit and numeric acceptance targets remain to establish through evaluation.	Team	Before final acceptance	Record connections
OQ-02	Which date should an issue timeline use? Resolved behavior. Supported development dates, preserved precision, reported/discussed fallback, uncertain ordering and planned/expected future dates are decided. Renderer details remain design work.	Team	Behavior resolved; implementation/inspection continues	Issue timelines

ID	Question	Owner	Decision point	Affected area
OQ-03	Which meetings have official minutes or transcripts that describe the outcomes? Resolved availability policy; source inspection ongoing. Availability is checked per selected meeting. Unpublished types need not block; failed expected inputs do. Cancelled meetings are accounted for with official evidence.	Team	Behavior resolved; implementation/inspection continues	Meeting overviews

Resolved behavior remains recorded under its original question identifier. Unresolved evaluation detail does not revoke the accepted policies or create additional product scope.

13 AI Assistance and Provenance

Table 21. AI assistance and provenance

Provenance item	Team record
Record	Version 1.0 review and revision
Tool and version	Codex
Dates and contributors	October 26–30, 2026. Kazim Yigit Ablay with AI assistance; v1.0 completed October 30.
Assistance provided	Helped organize and refine requirements, check consistency and traceability, and review references and formatting.
Inputs and sensitive data	Submitted SRS, project pitch, course template/worked example, selected-source feasibility artifacts and approved project requirements.
Human decisions	Kazim reviewed the proposed additions, accepted the functional/data/quality direction, revised policy scope and wording, selected v1.0 and authorized document update. Michael retains the lead on detailed search design.
Validation performed	AI-assisted source and document checks covered requirement mapping, references, source preservation and rendering. No product runtime testing or new stakeholder study was performed by this revision activity.
Changes or rejected output	Removed the narrow SQL-change ADR policy and personal repository-housekeeping policy; revised reference-evaluation and portability wording; retained categorization as conditional and excluded unagreed models, numerical targets and workflows.
Known limitations	Team review of this revision remains pending. Detailed semantic rubrics, aggregate acceptance targets, whole-meeting capacity and recovery/concurrency implementation require further evidence.
Record	Historical assistance record, October 6, 2026
Tool and version	Codex
Dates and contributors	October 6, 2026. Michael Avila with AI assistance.
Assistance provided	Reviewed the SRS against the supplied template and grading rubric; suggested wording for the context diagram, evidence register, misuse cases, prototype plan, quality requirements, traceability matrix, and user stories.
Inputs and sensitive data	Working SRS drafts, project pitch, course SRS template, and grading rubric.

Provenance item	Team record
Human decisions	Michael selected and edited suggestions for inclusion in the working draft. Joint approval remains pending.
Validation performed	AI-assisted review compared requirement sources with the supplied project pitch and checked document consistency and traceability. Michael reviewed suggestions during revision. No implementation testing or stakeholder validation was completed through this assistance.
Changes or rejected output	Revised source mappings after checking the pitch; distinguished directly supported requirements from derived requirements and pending team decisions.
Known limitations	Some requirement sources and acceptance criteria remain unresolved. Proposed additions require joint review, and verification activities remain planned.
Record	Historical initial-draft assistance record, October 5, 2026
Tool and version	Codex
Dates and contributors	October 5, 2026. Draft preparation by Kazim Yigit Ablay with AI assistance.
Assistance provided	Document organization, wording and formatting.
Inputs and sensitive data	Project proposal, speaker notes, course materials, SRS template and worked example.
Human decisions	Kazim selected the initial requirement wording and clarified the project scope.
Validation performed	Kazim reviewed the initial draft for consistency with the intended scope and corrected unsupported additions. Joint review and stakeholder validation were pending for that draft.
Changes or rejected output	Removed unagreed requirements, numerical targets, workflow details, unsupported evidence entries and AI-invented restrictions. Simplified wording and left unresolved details open.
Known limitations	The initial draft required further requirements and acceptance-criteria completion.

The team is responsible for understanding and defending the requirements and reviewing AI-assisted work. Historical assistance records describe their earlier revision state; they do not establish current approval or completed product validation.

Appendix A Requirements Change Log

The entries below record the changes incorporated from v0.2 into this v1.0 draft at Kazim Yigit Ablay's direction. Include in the Decision column means included in this draft; team review remains pending. Changes to acceptance or traceability are distinguished from changes to requirement wording.

Table 22. Requirements change log

ID	Date	Affected IDs	Change	Impact analysis	Decision
CH-01	Oct 30, 2026	Section 5.3, Table 1, All current requirement records	Set current document / requirement metadata to v1.0 and Draft for team review; retain historical versions.	Removes stale 0.1 / 0.2 current-status wording without claiming submission, completed testing or Michael's approval.	Included; review pending
CH-02	Oct 30, 2026	LIM-02	Express the existing microservices / MVC constraint as a binding requirement.	Preserves the selected architecture; responsibility detail moves to perspective / allocation and design notes.	Included; review pending
CH-03	Oct 30, 2026	LIM-03	Refine the web-frontend constraint to include React, TypeScript and Vite.	Makes the selected frontend core explicit; adds no routing, query-cache, UI-library or mobile-support commitment.	Included; review pending
CH-04	Oct 30, 2026	IF-02	Describe owner-controlled application APIs alongside the official record-source interface.	Connects data ownership and service integration; portal endpoint and long-operation details remain open.	Included; review pending
CH-05	Oct 30, 2026	IF-04	Add HTTP / JSON structured inter-service communication with binary / streamed document transfer permitted.	Clarifies communication checks without inventing transport-security versions or a callback protocol.	Included; review pending
CH-06	Oct 30, 2026	FR-01	Include official meeting-cancellation notices and link detailed input / preparation behavior to the new child requirements.	Extends official-record handling to the agreed cancellation case; no fixed agenda / minutes / transcript bundle or generated transcription.	Included; review pending
CH-07	Oct 30, 2026	FR-02	Expand search scope to issues, individual records and meetings; add selectable-combination requirements.	Preserves the search purpose and leaves matching / ranking details to Michael.	Included; review pending
CH-08	Oct 30, 2026	FR-03	Clarify source-supported issue connections; detail inferred inclusion, visible relationship basis, passage access and abstention in child requirements.	Preserves evidence / uncertainty obligations through FR-21 to FR-24 rather than inventing a confidence score or reviewer workflow.	Included; review pending
CH-09	Oct 30, 2026	FR-04	Clarify that the timeline presents the issue's supported evolution and link the date / planning refinements.	Preserves related-record access while distinguishing issue history from a chronological post feed.	Included; review pending

ID	Date	Affected IDs	Change	Impact analysis	Decision
CH-10	Oct 30, 2026	FR-05	Retain the statement and refine its acceptance / verification for recorded outcomes, unavailable information and cancellation.	Clarifies meaning through FR-38, NFR-01 / NFR-06 and cancellation checks; no assumed held-meeting actions.	Included; review pending
CH-11	Oct 30, 2026	FR-06	Retain the plain-language summary statement and extend quality / update traceability.	Adds material-qualification, factual-status, later-source and contradiction checks without a summary-length or accuracy percentage.	Included; review pending
CH-12	Oct 30, 2026	FR-07	Use event-driven source-link wording and extend verification to retained source versions and conflicting sources.	Preserves original-source navigation; passage access and provenance remain separately identified.	Included; review pending
CH-13	Oct 30, 2026	FR-08	Express official-video availability as an explicit EARS condition.	Preserves links-only behavior; no video analysis or new transcript generation.	Included; review pending
CH-14	Oct 30, 2026	DR-01	Retain the source-identification statement and extend data / provenance verification through DR-02 to DR-17.	Adds explicit original, relationship, date, evidence, version and recovery data rules without imposing an unagreed schema.	Included; review pending
CH-15	Oct 30, 2026	NFR-01	Use one binding statement covering factual claims in summaries and meeting overviews.	Preserves factual integrity and adds source-based quality checks; semantic rubric and aggregate acceptance targets remain pending.	Included; review pending
CH-16	Oct 30, 2026	NFR-02	Use event-driven safe-display wording; trace related model-input protection separately.	Preserves embedded-script prevention; does not claim prompt-only injection immunity.	Included; review pending
CH-17	Oct 30, 2026	NFR-03	Retain keyboard-operation wording and extend search verification to the three agreed search types.	Preserves accessibility scope without asserting full WCAG compliance or all-interface coverage.	Included; review pending
CH-18	Oct 30, 2026	NFR-04	Extend failure preservation from imported records / source references to successfully saved completed work for recovery.	Preserves previously saved records and completed work; does not imply an instant restart, fixed checkpoint granularity or public partial results.	Included; review pending
CH-19	Oct 30, 2026	COM-01	Retain course-tailored requirements guidance and add explicit language / change-control / traceability policy children.	Makes existing process expectations checkable without claiming formal ISO certification.	Included; review pending
CH-20	Oct 30, 2026	COM-02	Retain AI-provenance policy and add the current revision's assistance, human decisions and review limitations to section 13.	Preserves historical records and does not mislabel AI-assisted review as stakeholder validation or implementation testing.	Included; review pending

ID	Date	Affected IDs	Change	Impact analysis	Decision
CH-21	Oct 30, 2026	LIM-04–LIM-11, FR-09–FR-42, DR-02–DR-17, NFR-05–NFR-17, COM-03–COM-13	Add the mapped design, behavior, data, quality and policy entries.	Adds 82 new identifiers; preserves categorization as conditional and keeps unselected technologies / targets out.	Included; review pending
CH-22	Oct 30, 2026	MU-01, MU-02, MU-03, MU-04, MU-05, MU-06	Repair misuse-case traces and add the imported-model-instruction scenario.	Removes semantically invalid legacy FR-10 / FR-11 references before those numbers are assigned to new requirements.	Included; review pending
CH-23	Oct 30, 2026	US-01, US-02, US-03	Retain story identifiers and prepare separate, testable acceptance scenarios linked to the refined requirements.	Avoids the old combined search / navigation criterion and clarifies evidence / date / planned-event behavior.	Included; review pending
CH-24	Oct 30, 2026	OQ-01, OQ-02, OQ-03	Record resolved behavioral decisions while keeping evidence-rubric / measurement work explicit.	Prevents stale open questions from making settled scope appear undecided; no new expanding question list.	Included; review pending
CH-25	Oct 30, 2026	Section 5.1, Section 5.3, Section 7 introduction, Section 11, V-01–V-28	Update requirement language / attribute text and planned verification / traceability for the v1.0 set.	Preserves V-01 to V-15 meanings and reserves V-16 to V-28 as planned scenarios, not executed test artifacts.	Included; review pending
CH-26	Oct 30, 2026	EV-03, PR-01, Section 10, Section 13	Align selected-record feasibility evidence, allocation and revision provenance with actual supported status.	Do not invent broader quality validation, PDR feedback, participant outcomes, human production assignments or release commitments.	Included; review pending
CH-27	Oct 30, 2026	Section 1.2, Section 1.3, Section 1.6, Section 2.1, Section 2.6	Align scope, definitions and operating context with the agreed issue / development / evidence model, current reconstructed history and selected architecture.	Preserves existing source / feature exclusions, keeps non-English processing nonmandatory and keeps detailed implementation in design guidance; no new coverage guarantee or production hosting choice.	Included; review pending